Chapter 2: Quadratics

Chapter 2 · Original slide 1 · Quadratics

Pure Y1 Ch 2
Quadratics

Chapter 2 · Original slide 2 · Chapter Overview

Again, most of the content in this chapter is not new, but brings together a variety of algebraic and graph sketching GCSE skills regarding quadratic equations.

1: Solving quadratic equations

2: Completing the square

Solve
$(x+1)^2 - 3(x+1) + 2 = 0$

Write $2x^2 + 8x - 5$ in the form $a(x+b)^2 + c$

4: Quadratic Graphs

If $f(x) = x^2 + 2x$, find the roots of $f(x)$.

3: Quadratics as functions

Sketch $y = x^2 + 4x - 5$, indicating the coordinate of the turning point and any intercepts with the axes.

Find the range of values of k for which $x^2 + 4x + k = 0$ has two distinct real solutions.

5: The Discriminant

6: Modelling with Quadratics

Chapter 2: Quadratics

Chapter 2 · Original slide 3 · Why do we care about quadratics…

Summations





Projectile Motion

A projected object, acting only under gravity, follows a *parabolic* trajectory, i.e. its path can be described using a quadratic equation.

In Mechanics you will see the formula:

$s = ut + \frac{1}{2}at^2$

Where s is distance/displacement, u is initial speed/velocity, t is time and a is acceleration (in this case gravitational acceleration).

This equation is quadratic in t.

Good question!

There are a number of practical scenarios where a quadratic relationship arises between variables...

The sum of 1 to n is $\frac{1}{2}n^2 + \frac{1}{2}n$. I recently had to make a Sports Day spreadsheet which took the number of competing school 'houses' and worked out the total points allocated for an event by finding the formula of the resulting quadratic sequence.

1 competitor: **8** points

2 competitors: $\mathbf{8+7=15}$

3 competitors: $\mathbf{8+7+6=21}$

$\mathbf{n}$ competitors: $\mathbf{\frac{17}{2}n-\frac{1}{2}n^2}$

Chapter 2 · Original slide 4 · Why do we care about quadratics…

Hannah's sweets

19 There are n sweets in a bag. 6 of the sweets are orange. The rest of the sweets are yellow. Hannah takes at random a sweet from the bag. She eats the sweet. Hannah then takes at random another sweet from the bag. She eats the sweet. The probability that Hannah eats two orange sweets is $\frac{1}{3}$. (a) Show that $n^2 - n - 90 = 0$.

Quadratic expressions also regularly emerge when there's a product of two expressions involving the same variable (in this case because Hannah ate two orange sweets).

Chapter 2: Quadratics

Chapter 2 · Original slide 5 · 1: Solving Quadratic Equations

$x^2 + 5x = 6$

There are three ways of solving a quadratic equation.

One is by *completing the square*, which we'll do later.

What are the other two ways?

By factorisation

Using the quadratic formula.

If $ax^2 + bx + c = 0$ then

$x^2 + 5x - 6 = 0$

Put in form.

$ax^2 + bx + c = 0$ with 0 on one side

Factorise. If the product of two things is 0, at least one must be 0.

Chapter 2 · Original slide 6 · 1: Solving Quadratic Equations

Solving without factorising

If the subject only appears once however, it might be easier not to expand out/factorise:

If you can't see why the $\pm$ is required, think about the solutions to: $x^2 = 4$.

$2^2 = 4$, but $(-2)^2 = 4$ as well!

So $x = \pm 2$.

Quadratics 'in disguise'

When we have an expression like say $x^2 + 3x - 2$, we say it is "quadratic in x". In trigonometry you will have to solve equations like $(\sin x)^2 + 3\sin x + 2$. We say that the expression is "quadratic in $\sin x$".

Either use a suitable substitution so that you have a 'normal' quadratic, or go straight for the factorisation if you're feeling more confident (recommended!).

Solve $x - 6\sqrt{x} + 8 = 0$

substitution

hardcore

Chapter 2: Quadratics

Chapter 2 · Original slide 7 · Test Your Understanding

1
Solve $(x+3)^2 = x+5$ using factorisation.

2
Solve $(2x+1)^2 = 5$

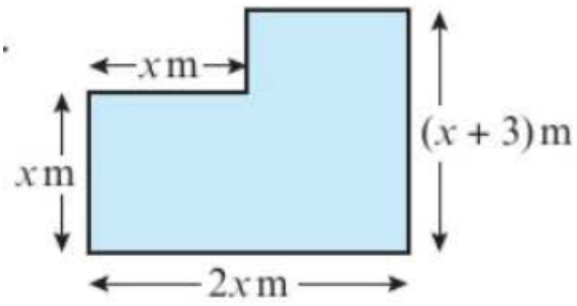
3
Solve $\sqrt{x+3} = x-3$

4
Solve $2x + \sqrt{x} - 1 = 0$

"Use of Technology":

The 'equation' mode on your calculator will solve quadratics in the form $ax^2 + bx + c = 0$. When you're asked for the 'order', use 2 (we'll see why later).

Chapter 2 · Original slide 8 · Exercise 2A/2B Pure Y1 Pages 20-22



3 Solve the following equations:

i $6x^2 - 7 = 11x$

h $5x^2 - 10x^2 = -7 + x + x^2$

4 This shape has an area of $44\,\text{m}^2$. Find the value of x.

5 Solve the equation $5x + 3 = \sqrt{3x+7}$.

Chapter 2 · Original slide 9 · Exercise 2A/2B Pure Y1 Pages 20-22

2 Solve the following equations using the quadratic formula. Give your answers to three significant figures.

h $2x^2 + 11x = 5x^2 - 18$

3 For each of the equations below, choose a suitable method and find all of the solutions. Where necessary, give your answers to three significant figures.

e $(2x + 8)^2 = 100$ **f** $6x^2 + 6 = 12x$

Chapter 2 · Original slide 10 · Exercise 2A/2B Pure Y1 Pages 20-22

1

Extension: (Full Database: http://www.drfrostmaths.com/resources/resource.php?rid=268)

Chapter 2: Quadratics

Chapter 2 · Original slide 11 · 2: Completing the Square

“Completing the square” means putting a quadratic in the form $(x+a)^2+b$

or $a(x+b)^2+c$

The underlying reason we do this is because $\mathbf{x}$ only appears once in the expression (e.g. in $(x+2)^2+3$ vs x^2+4x+7), which makes it algebraically easier to handle. This has a number of consequences:

a. Solving Quadratics

If we have a completed square:

$(x+4)^2-7=0$

we saw at the start of the chapter how we could rearrange to make x the subject.

Indeed using the quadratic formula is actually solving the quadratic by completing the square – it’s just someone has done the work for us already!

b. Sketching Quadratics

We’ll see later that if $y=(x+a)^2+b$, then the minimum point is $(-a,b)$

c. In integration

In Further Maths, completing the square allows us to ‘integrate’ expressions like:

$\int \frac{1}{x^2-4x+5}dx$

(you will cover integration later this module)

Chapter 2 · Original slide 12 · Completing the Square Recap

Textbook Note:

The textbook uses the formula

$x^2+bx=\left(x+\frac{b}{2}\right)^2-\left(\frac{b}{2}\right)^2$

and similarly

$ax^2+bx+c=a\left(x+\frac{b}{2a}\right)^2+\left(c-\frac{b^2}{4a}\right)$

My personal judgement is that it’s not worth memorising these and you should instead think through the steps. Even the textbook agrees!

Expand:

$(x+\mathbf{9})^2=$
$(x-\mathbf{5})^2=$

What do you notice about the relationship between the bold numbers?

Therefore if we had x^2+12x, how could we write it in the form $(x+a)^2+b$?

Since $(x+6)^2=x^2+12x+36$, we want to discard the 36, so ‘throw it away’ by subtracting.

Further Examples:

$x^2+8x=$
$x^2-2x=$
$x^2-6x+7=$

Notice that despite the a being negative, we still subtract after the bracket as $(-1)^2$ is positive.

Chapter 2: Quadratics

Chapter 2 · Original slide 13 · Completing the Square

Express $2x^2 + 12x + 7$ in the form $a(x + b)^2 + c$

Factorise out coefficient of x^2. You can leave the constant term outside the bracket.

Complete the square inside the bracket (you should have two sets of brackets)

Expand out outer bracket.

Express $5 - 3x^2 + 6x$ in the form $a - b(x + c)^2$

It may help to write in the form $ax^2 + bx + c$ first.

Chapter 2 · Original slide 14 · Test Your Understanding

Express $3x^2 - 18x + 4$ in the form $a(x + b)^2 + c$

Express $20x - 5x^2 + 3$ in the form $a - b(x + c)^2$

Chapter 2 · Original slide 15 · Solving by Completing the Square

Solve the equation:

$3x^2 - 18x + 4 = 0$

Reflections: Previously we factorised out the 3. This is because $3x^2 - 18x + 4$ on its own is an expression, so dividing by 3 (instead of factorising) would change the expression.

However, in an equation, we can divide both sides by 3 without affecting the solutions.

Chapter 2 · Original slide 16 · Exercise 2C/2D Pure Y1 Page 23-24

1 Complete the square for the expressions:

a $x^2 + 4x$
b $x^2 - 6x$
c $x^2 - 16x$
d $x^2 + x$
e $x^2 - 14$

2 Complete the square for the expressions:

a $2x^2 + 16x$
b $3x^2 - 24x$
c $5x^2 + 20x$
d $2x^2 - 5x$
e $8x - 2x^2$

Chapter 2 · Original slide 17 · Exercise 2C/2D Pure Y1 Page 23-24

4 Given that $x^2 + 3x + 6 = (x + a)^2 + b$ find the values of the constants a and b

5 Write $2 + 0.8\text{x} - 0.04x^2$ in the form $A - B(x + C)^2$ where $A \text{and} B$are constants to be determined

4 By completing the square, show that the solutions to the equation $x^2 + 2bx + c = 0$ are given by the formula $x = -b \pm \sqrt{b^2 - c}$. (4 marks)

Chapter 2 · Original slide 18 · Proving the Quadratic Formula

If $ax^2 + bx + c = 0$, prove that $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Just use exactly the same method as you usually would!

Chapter 2 · Original slide 19 · 3: Functions

A function is something which provides a rule on how to map inputs to outputs.

We saw at GCSE that functions were a formal way of describing a ‘number machine’:

Input

Output

x

f

$2x$

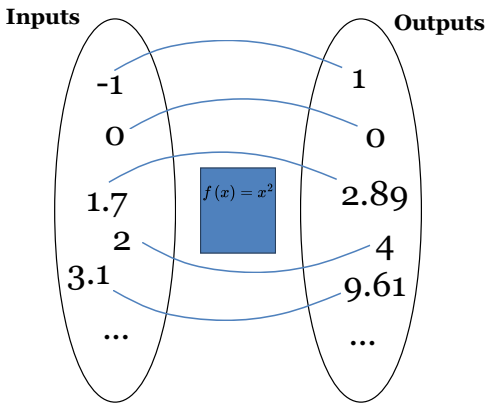
Name of the function (often f or g)

Input

Output

$f(x) = 2x$

Chapter 2 · Original slide 20 · 3: Functions



You’ll cover functions extensively in future chapters, but for now, you need to understand the following concepts:

! The roots/zeroes of a function are the values of x for which $f(x) = 0$.

! The domain of a function is the set of possible inputs.

! The range of a function is the set of possible outputs.

Chapter 2 · Original slide 21 · Examples

If $f(x) = x^2 - 3x$ and $g(x) = x + 5, x \in \mathbb{R}$

a) Find $f(-4)$

b) Find the values of x for which $f(x) = g(x)$

c) Find the roots of $f(x)$.

d) Find the roots of $g(x)$.

Note: The domain is usually stated for you.

c d

The roots are the inputs which give an output of 0. So set output expression to 0.

a b

Conceptually, we're looking for the inputs of the functions which give the same outputs. We can just equate the output expressions.

Chapter 2 · Original slide 22 · Examples

Determine the minimum value of the function $f(x) = x^2 - 6x + 2$, and state the value of x for which this minimum occurs.

This means we want to minimise the output of the function.

You might try a (clumsy) approach of trying a few values of x and try to see what makes the output as small as possible...

But the best way to find the minimum/maximum value of a quadratic is to complete the square:

Since anything squared is at least 0, the smallest we can make the bracket is 0.

! If $f(x) = (x + a)^2 + b$, the minimum value of $f(x)$ is b, which occurs when $x = -a$.

Chapter 2: Quadratics

Chapter 2 · Original slide 23 · Quickfire Questions

$\mathbf{f(x)}$	Completed square	Min/ max value of $\mathbf{f(x)}$	$\mathbf{x}$ for which this min/max occurs
$x^2 + 4x + 9$			
$x^2 - 10x + 21$			
$10 - x^2$			
$8 - x^2 + 6x$			

Chapter 2 · Original slide 24 · Test Your Understanding

1

Find the minimum value of $f(x) = 2x^2 + 12x - 5$ and state the value of x for which this occurs.

2

Find the roots of the function $f(x) = 2x^2 + 3x + 1$

3

Find the roots of the function $f(x) = x^4 - x^2 - 6$

Chapter 2 · Original slide 25 · Exercise 2E Pure Y1 Pages 26-27

2 The function $\mathrm{f}(x)$ is defined by $\mathrm{f}(x) = x^2 - 2x,\ x \in \mathbb{R}$. Given that $\mathrm{f}(a) = 8$, find the two possible values for a.

6 The function f is defined as $\mathrm{f}(x) = x^2 - 2x + 2,\ x \in \mathbb{R}$.

a Write $\mathrm{f}(x)$ in the form $(x + p)^2 + q$, where p and q are constants to be found.
b Hence, or otherwise, explain why $\mathrm{f}(x) > 0$ for all values of x, and find the minimum value of $\mathrm{f}(x)$.

8 The function f is defined as $\mathrm{f}(x) = 3^{2x} - 28(3^x) + 27,\ x \in \mathbb{R}$.

a Write $\mathrm{f}(x)$ in the form $(3^x - a)(3^x - b)$, where a and b are real constants.
b Hence find the two roots of $\mathrm{f}(x)$.

Chapter 2 · Original slide 26 · 4: Quadratic Graphs

Recall that x refers to the input of a function, and the expression $f(x)$ refers to the output. For graph sketches, we often write $y = f(x)$, i.e. we set the y values to be the output of the function.

y

A ‘sketch’ in maths has a clearly defined meaning:

• We care about the general shape of the graph, not exact points.

• The axes should have no scale.

• Generally the only coordinates indicated are intercepts with the axis or other points of interest (e.g. intersections of multiple graphs)

Features needed in sketch of quadratic?

Recall a root of a function is where the output, in this case the y value, is 0.

Roots

x

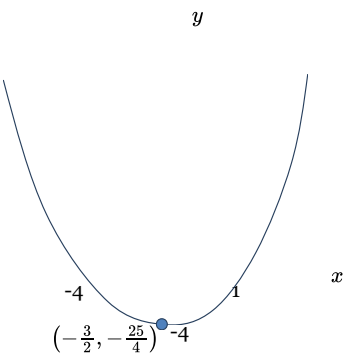
General shape: Smiley face or hill?

y-intercept

Turning point

Chapter 2: Quadratics

Chapter 2 · Original slide 27 · Example



Sketch the graph of $y = x^2 + 3x - 4$ and find the coordinates of the turning point.

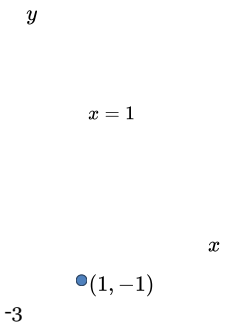
Roots:

y-intercept:

Turning point:

Recall that if $f(x) = (x + a)^2 + b$, the minimum output is b and $-a$ is the x value which minimises it. i.e. Turning point is $(-a, b)$

Chapter 2 · Original slide 28 · Example



Sketch the graph of $y = 4x - 2x^2 - 3$ and find the coordinates of the turning point. Write down the equation of the line of symmetry.

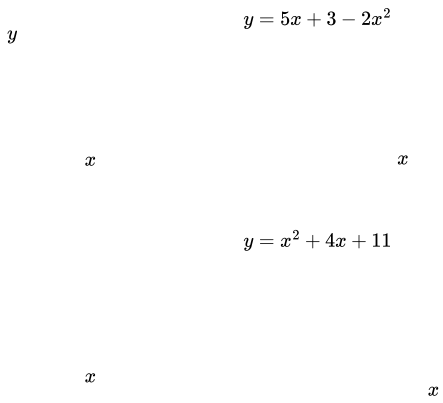
Roots:

y-intercept:

Turning point:

Line of reflection:

Chapter 2 · Original slide 29 · Test Your Understanding



Sketch the following, indicating any intercepts with the axis, the turning point and the equation of the line of symmetry.

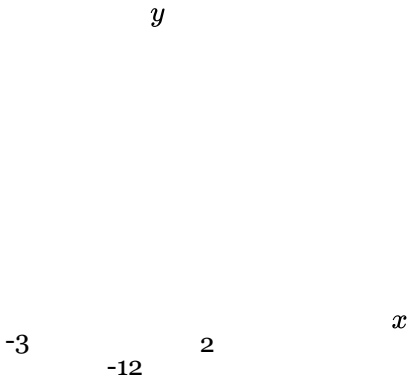
a c

$y = x^2 + 4$

b d

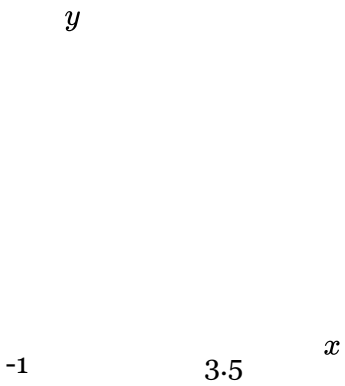
$y = x^2 - 7x + 10$

Chapter 2 · Original slide 30 · Determining the Equation using a Graph



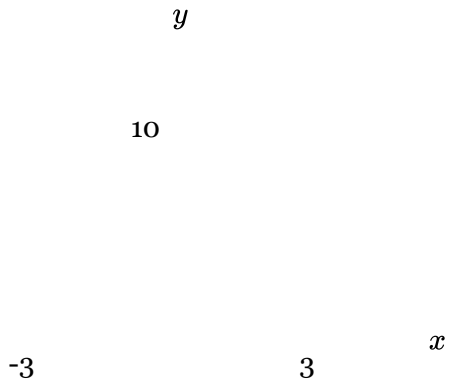
Determine the equation of this quadratic graph, in the form $y = ax^2 + bx + c$.

Chapter 2 · Original slide 31 · Further Example



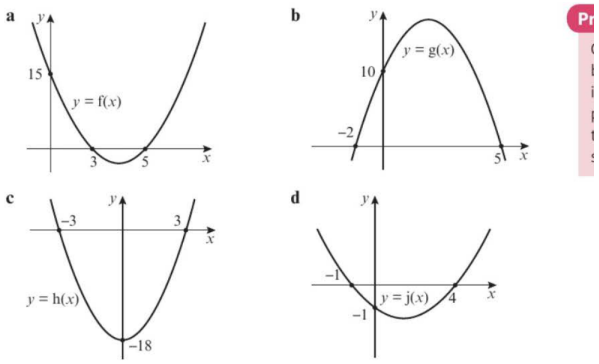
Determine the equation of this quadratic graph, in the form $y = ax^2 + bx + c$, where a, b, c are integers.

Chapter 2 · Original slide 32 · Test Your Understanding



Determine an equation of this quadratic graph.

Chapter 2 · Original slide 33 · Exercise 2F Pure Y1 Page 30



2 These sketches are graphs of quadratic functions of the form $ax^2 + bx + c$. Find the values of a,b,c for each function.

Chapter 2 · Original slide 34 · Exercise 2F Pure Y1 Page 30

3 The graph of $y = ax^2 + bx + c$ has a minimum at (5,-3) and passes through (4,0). Find the values of a,b,c.

[MAT 2003 1H] Into how many regions is the plane divided when the following three parabolas are drawn?

$y = x^2$
$y = x^2 - 2x$
$y = x^2 + 2x + 2$

x

Extension Question:

y

Chapter 2 · Original slide 35 · Starter

How many distinct real solutions do each of the following have?

$x^2 - 12x + 36 = 0$

$x^2 + x + 3 = 0$

$x^2 - 2x - 1 = 0$

Chapter 2 · Original slide 36 · 5: The Discriminant

$ax^2 + bx + c = 0$

$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Note: Roots of a function f are the values of x such that $f(x) = 0$.

Similarly the roots of an equation are solutions to an equation in the form $f(x) = 0$

Looking at this formula, when in general do you think we have:

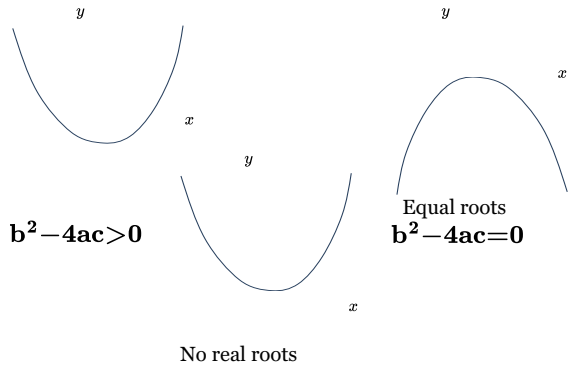
- No real roots?
- Equal roots?
- Two distinct roots?

$b2 - 4ac$is known as the discriminant.

Because adding 0 or subtracting 0 in the quadratic formula gives the same value.

Chapter 2: Quadratics

Chapter 2 · Original slide 37 · 5: The Discriminant



We say the graph “touches” the axis.

Distinct real roots

$b^2-4ac<0$

Chapter 2 · Original slide 38 · Just for your interest…

Why do we say “Equal Roots” not “One root”?

It is due to the Fundamental Theorem of Algebra:

“Every polynomial of order n has exactly n roots.”

A polynomial is an expression with non-negative integer powers of x, i.e.

$a + bx + cx^2 + dx^3 + \ldots$

All linear, quadratic and cubic expressions are examples of polynomials.

$x^2 - 12x + 36 = 0$

Using the quadratic formula gives us the same value in both $+$ and $-$ cases: $x = 6$.

You might wonder why we say “*it has one repeated root*” or “*it has equal roots*”, i.e. indicating we have 2 roots (but with the same value). Why not say it has 1 root?

Despite the theorem being a simple statement, it was only until 1806 that it was first proven by Argand. Clearly by using the quadratic formula we can show a quadratic equation has 2 roots. We can use similar formulas to show that a cubic has 3 roots and a quartic 4 roots. But there is provably no such formulae for order 5 (quintics) and beyond. So we have to prove for example that 5 roots exist for a quintic, despite us having no way to find these exact roots!

One side result of the Fundamental Theorem of Algebra is that every polynomial can be written as a product of linear and/or quadratic expressions.

Leibniz claimed in 1702 that a polynomial of the form $x^4 + a^4$ cannot be written in this way. He then got completely burned by Euler in 1742 who managed to do so:

$$x^4 + a^4 = \left(x^2 + a\sqrt{2}x + a^2\right)\left(x^2 - a\sqrt{2}x + a^2\right)$$

The order of a polynomial is its highest power of x. So the order of a quadratic is 2, and a cubic 3.

Chapter 2: Quadratics

Chapter 2 · Original slide 39 · Quickfire Questions

Equation	Number of Distinct Real Roots
$x^2 - 4x + 1 = 0$	
$x^2 - 4x + 4 = 0$	
$2x^2 - 6x - 3 = 0$	
$x - 4 - 3x^2 = 0$	
$1 - x^2 = 0$	

Equation	Discriminant
$x^2 + 3x + 4 = 0$	

Chapter 2 · Original slide 40 · Problems involving the discriminant

8. The equation $x^2 + 2px + (3p + 4) = 0$, where p is a positive constant, has equal roots.

(a) Find the value of p. (4)

(b) For this value of p, solve the equation $x^2 + 2px + (3p + 4) = 0$. (2)

a)

Tip: Always start by writing out a, b and c explicitly.

b)

Chapter 2 · Original slide 41 · Exercise 2G Pure Y1 Page 32

2 Find the values of k for which $x^2 + 6x + k = 0$ has two real solutions.

3 Find the value of t for which $2x^2 - 3x + t = 0$ has exactly one solution.

Chapter 2 · Original slide 42 · Exercise 2G Pure Y1 Page 32

4 Given that the function $\mathrm{f}(x) = sx^2 + 8x + s$ has equal roots, find the value of the positive constant s.

5 Find the range of values of k for which $3x^2 - 4x + k = 0$ has no real solutions.

Chapter 2 · Original slide 43 · Exercise 2G Pure Y1 Page 32

7 $h(x) = 2x^2 + (k + 4)x + k$, where k is a real constant.

a Find the discriminant of $h(x)$ in terms of k.

b Hence or otherwise, prove that $h(x)$ has two distinct real roots for all values of k.

6 The function $g(x) = x^2 + 3px + (14p - 3)$, where p is an integer, has two equal roots.

a Find the value of p.

b For this value of p, solve the equation $x^2 + 3px + (14p - 3) = 0$.

Chapter 2 · Original slide 44 · Test Your Understanding

$x^2 + 5kx + (10k + 5) = 0$

where k is a constant.

Given that this equation has equal roots, determine the value of k.

Find the range of values of k for which $x^2 + 6x + k = 0$ has two distinct real solutions.

Chapter 2 · Original slide 45 · Exercise 2G Pure Y1 Page 32

Extension Questions:

1

[MAT 2009 1C] Given a real constant c, the equation

$x^4 = (x - c)^2$

Has four real solutions (including possible repeated roots) for:

A) $c \leq \frac{1}{4}$

B) $-\frac{1}{4} \leq c \leq \frac{1}{4}$

C) $c \leq -\frac{1}{4}$

D) all values of c

2

[MAT 2006 1B] The equation $\left(2 + x - x^2\right)^2 = 16$ has how many real root(s)?

3

[MAT 2011 1B] A rectangle has perimeter P and area A. The values P and A must satisfy:

A) $P^3 > A$

B) $A^2 > 2P + 1$

C) $P^2 \geq 16A$

D) $PA > A + P$

Chapter 2 · Original slide 46 · Modelling

The new A Level has a particular emphasis of the application of theory to real-life situations. A mathematical model is the maths used to model such a situation, possibly with some simplifying assumptions.

Example (from textbook): A spear is thrown over level ground from the top of a tower.

The height, in metres, of the spear above the ground after t seconds is modelled by the function:
$h(t) = 12.25 + 14.7t - 4.9t^2, t \geq 0$

a) Interpret the meaning of the constant term 12.25 in the model.

b) After how many seconds does the spear hit the ground?

c) Write $h(t)$ in the form $A - B(t - C)^2$, where A, B and C are constants to be found.

d) Using your answer to part c or otherwise, find the maximum height of the spear above the ground, and the time at which this maximum height is reached?

Chapter 2 · Original slide 47 · Exercise 2H Pure Y1 Page 34

2 A car manufacturer uses a model to predict the fuel consumption, y miles per gallon (mpg), for a specific model of car travelling at a speed of x mph.

$$y = -0.01x^2 + 0.975x + 16,\ x > 0.$$

a Use the model to find two speeds at which the car has a fuel consumption of 32.5 mpg.
b Rewrite y in the form $A - B(x - C)^2$, where A, B, C are constants to be found.
c Using your answer to part b, find the speed at which the car has the greatest fuel efficiency.
d Use the model to calculate the fuel consumption of a car travelling at 120 mph.
Comment on the validity of using this model for very high speeds.

Chapter 2 · Original slide 48 · Exercise 2H Pure Y1 Page 34

3 A fertiliser company uses a model to determine how the amount of fertiliser used, f kilograms per hectare, affects the grain yield g, measured in tonnes per hectare.

$$g = 6 + 0.03f - 0.00006f^2.$$

a According to the model, how much grain would each hectare yield without any fertiliser? (1 mark)
b One farmer currently uses 20 kilograms of fertiliser per hectare. How much more fertiliser would he need to use to increase his grain yield by 1 tonne per hectare? (4 marks)

Chapter 2 · Original slide 49 · Exercise 2H Pure Y1 Page 34

4 A football stadium has 25000 seats. The football club know from past experience that they will sell only 10000 tickets if each ticket costs \pounds30. They also expect to sell 1000 more tickets every time the price goes down by \pounds1.

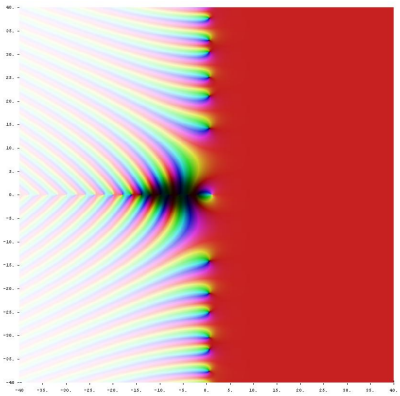
a The number of tickets sold t can be modelled by the linear equation $t = M - 1000p$, where \pounds$p$ is the price of each ticket and M is a constant. Find the value of M. (1 mark)

The total revenue, \pounds$r$, can be calculated by multiplying the number of tickets sold by the price of each ticket. This can be written as $r = p(M - 1000p)$.

b Rearrange r into the form $A - B(p - C)^2$, where A, B, C are constants to be found. (3 marks)

c Using your answer to part b or otherwise, work out how much the football club should charge for each ticket if they want to make the maximum amount of money. (2 marks)

Chapter 2 · Original slide 50 · Would you like $1,000,000 for finding roots?



The Riemann Zeta Function is a function that allows you to do the infinite sum of powers of reciprocals, e.g.

$\zeta(2) = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \ldots$
$\zeta(3) = \frac{1}{1^3} + \frac{1}{2^3} + \frac{1}{3^3} + \ldots$

One of the 8 'Clay Millennium Problems' (for which solving any attracts a $1,000,000 prize) is to showing all roots of this function have some particular form, i.e. the form of x such that $\zeta(x) = 0$.

We saw earlier that the roots of a function f are the values x such that $f(x) = 0$.